

Raghav Senthil Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Worked with an *Agile* team of **10 engineers** to integrate and deploy a cloud-hosted *Python (FastAPI)* search service on *AWS*, running in production for a high school debate-research SaaS platform with **10K+ site visitors**
- Tuned *Elasticsearch* indexing across **600K+ documents** for scalability, answering queries in **under 1 second**
- Cleaned the search index by flagging the **30%** of documents that duplicated others with a de-duplication algorithm

Product Development Intern | Jetson

January 2025 – February 2025

- Evaluated **3 experimental features** across **7 UI/UX test iterations** on the Jetson App (**100K+ users**)
- Wrote up findings as a **5-page report** of engagement-optimizing recommendations delivered to Jetson's CEO
- Cut onboarding from **12 to 5 pages** in *Figma*, confirming the improvement with **A/B testing** on 30 test users

Data Analytics Intern | STEM E

June 2024 – July 2024

- Prepared and analyzed **3 years** of TikTok data (**200+ posts**) using *Python (pandas)*, *Power BI* and *Excel*
- Found out that science “fun facts” posts made up just **7%** of all posted content yet earned **69.3%** of all views
- Turned that finding into a “fun facts” content strategy the Executive Director adopted, lifting engagement **24%**

PROJECTS

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Led a team of **4** to design, build and deploy a VLM-based pricing algorithm for **Kamion (YC S22)** that addressed a complex business problem using generative AI models, earning **2nd Place at UIUC Founder's Hackathon**
- Cleaned and prepared data from **15,000** scraped truck-listing images via de-duplication and non-USD filtering
- Engineered a generative AI agent (*Claude Sonnet 4.6*) with a *JSON schema* to extract truck details from photos
- Tested and refined the extraction schema until **10 consecutive images** were correct on each of **3 features**
- Fit a *quantile regression* model on the extracted features to predict prices at **87% accuracy** across **30 samples**

EcoBin | [Research Paper](#) | [Source Code](#)

March 2026 – May 2026

- Designed, trained, tested and evaluated a two-stage *EfficientNetV2-S* deep learning neural network (**4.39M** parameters) in *PyTorch* on **15K+ images** to classify waste items into 4 disposal pathways at **96.13% accuracy**
- Prepared data for machine learning with a synthetic pipeline (*U2-Net*, alpha-compositing) generating **9K** images
- Shared the work as an *arXiv* paper and **2 Kaggle datasets**; **placed 2nd at Arizona State Science Fair**

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Deployed a **RAG** AI agent (*Claude Sonnet 4.5*) that uses vector search to retrieve session-scoped documents
- Pulled papers from *OpenAlex* and filled missing metadata from *Semantic Scholar* in a two-API data pipeline
- Queried a *PostgreSQL* database hosted on *Supabase* with *SQL* to analyze papers by field, year and citation count
- Rendered each paper's outbound citations as an interactive *React Flow* graph spanning **26 research fields**

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Integrated **4 Random Forest** machine learning models into a *Python FastAPI* back end that monitors live water, waste and energy data from **3 tools** to detect anomalies and deliver actionable steps via a *Gemma* Voice LLM
- Built a set of **3 edge AI tools** that track a school's energy, water and food consumption for anomaly detection
- Directed a team of **5** from initial ideation to the final presentation as a **USAII Global Hackathon Finalist**

SKILLS

Programming Languages: Python, SQL, Java, C++, JavaScript, HTML/CSS

Frameworks/Platforms: PostgreSQL, Supabase, AWS, Google Cloud, FastAPI, MySQL, Elasticsearch, Flask

AI/ML: PyTorch, TensorFlow, scikit-learn, Keras, Hugging Face, Claude API, OpenCV, Gemma 4, Grad-CAM

Data & Analytics: pandas, NumPy, A/B Testing, Power BI, Microsoft Excel, Hootsuite

Concepts/Practices: Model Training, Model Evaluation, Generative AI, Vector Search, Data Preparation